



"M1"

Newton's LAWS OF MOTION

✓ Lec'05

✓ Lec'06



M1

"LAWS OF MOTION"

LEC'05

✓ - ✓

$$F = ma$$

✓ INCLINED

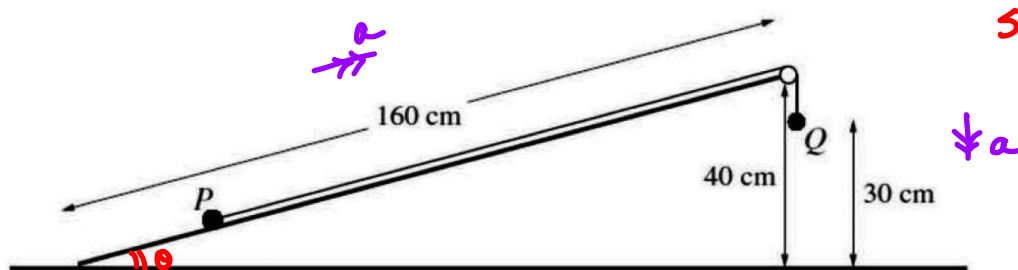
SMOOTH

ROUGH

✓ Resolution



Question 31:

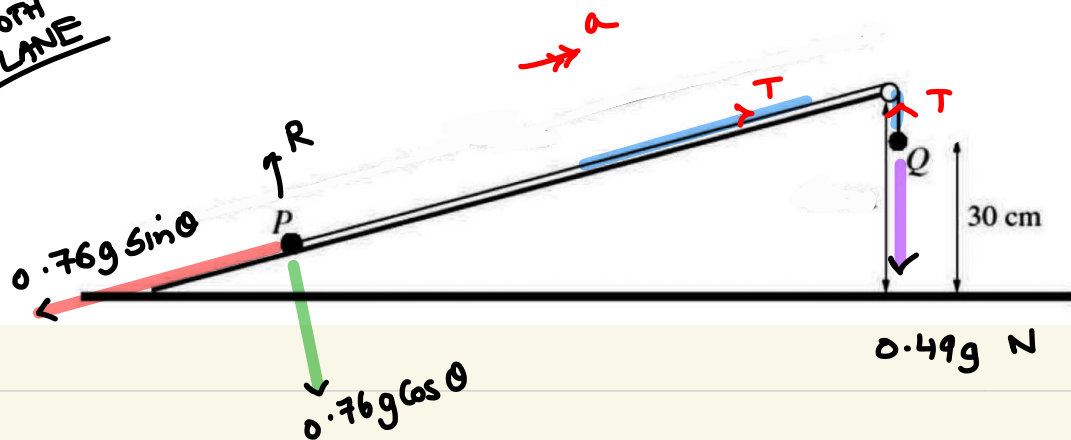


A smooth inclined plane of length 160 cm is fixed with one end at a height of 40 cm above the other end, which is on horizontal ground. Particles P and Q , of masses 0.76 kg and 0.49 kg respectively, are attached to the ends of a light inextensible string which passes over a small smooth pulley fixed at the top of the plane. Particle P is held at rest on the same line of greatest slope as the pulley and Q hangs vertically below the pulley at a height of 30 cm above the ground (see diagram). P is released from rest. It starts to move up the plane and does not reach the pulley. Find

- (i) the acceleration of the particles and the tension in the string before Q reaches the ground, [4]
- (ii) the speed with which Q reaches the ground, [2]
- (iii) the total distance travelled by P before it comes to instantaneous rest. [3]

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SMOOTH PLANE



$$\sin \theta = \frac{1}{4}$$

(i) $0.49g - T = 0.49a$

$$T - 0.76g \left(\frac{1}{4} \right) = 0.76a$$

$$0.49 - 0.19g = 1.25a$$

$$0.30g = 1.25a \quad \therefore a = 2.4 \text{ m/s}^2$$

$$T =$$

(ii) $u = 0, a = 2.4, s = 0.3 \text{ m}$

$$v^2 = 0^2 + 2(2.4)(0.3) \quad \therefore v = 1.2 \text{ m/s}$$

(iii) $T - \boxed{0.76}g \left(\frac{1}{4} \right) = \boxed{0.76}a \quad \therefore a = -2.5 \text{ m/s}^2$

$$u = 1.2, a = -2.5, v = 0$$

$$s = ?$$

$$0^2 = 1.2^2 + 2(-2.5)s \quad \therefore s = 0.288 \text{ m}$$

$$\text{Total dist} \rightarrow 0.288 + 0.30 = 0.588 \text{ m}$$

31 (i) For $T - (40 \div 160) \times 0.76g = 0.76a$ or
 $0.49g - T = 0.49a$

For $0.49g - T = 0.49a$ or
 $T - (40 \div 160) \times 0.76g = 0.76a$ or
 $0.49g - (400 \div 160) \times 0.76g =$
 $(0.49 + 0.76)a$

Acceleration is 2.4 ms^{-2}
and tension is 3.72 N (3.724 exact)

(ii) $[v^2 = 2 \times 2.4 \times 0.3]$

Speed is 1.20 ms^{-1}

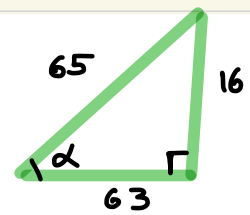
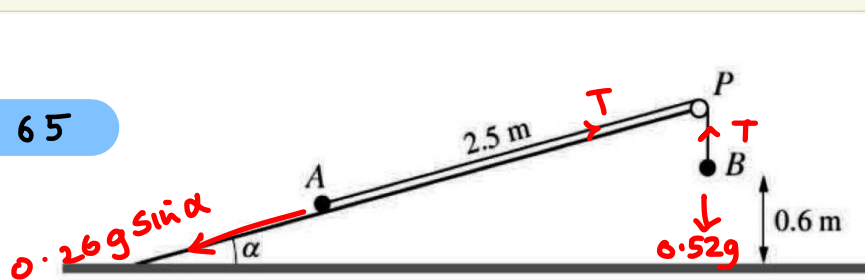
(iii) Distance while Q is on the ground
 $= (2 \times 2.4 \times 0.3) \div 2(40g \div 160)$
 $(= 0.288 \text{ m})$

Distance travelled is 0.588 m



Question 24:

16 - 63 - 65



Particles A of mass 0.26 kg and B of mass 0.52 kg are attached to the ends of a light inextensible string. The string passes over a small smooth pulley P which is fixed at the top of a smooth plane. The plane is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{16}{65}$ and $\cos \alpha = \frac{63}{65}$. A is held at rest at a point 2.5 metres from P , with the part AP of the string parallel to a line of greatest slope of the plane. B hangs freely below P at a point 0.6 m above the floor (see diagram). A is released and the particles start to move. Find

- (i) the magnitude of the acceleration of the particles and the tension in the string, [5]
- (ii) the speed with which B reaches the floor and the distance of A from P when A comes to instantaneous rest. [6]

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$$0.52g - T = 0.52a$$

$$T - 0.26g \left(\frac{16}{65} \right) = 0.26a$$

$$0.52g - 0.26g \left(\frac{16}{65} \right) = (0.52 + 0.26)a$$

$$a = 5.85 \text{ m/s}^2$$

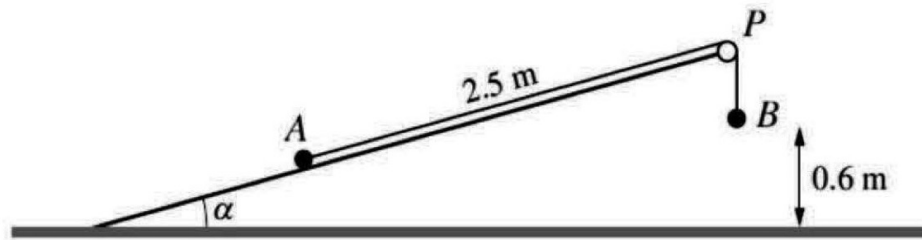
$$T =$$

$$u = 0, \quad s = 0.6, \quad a = 5.85$$

$$v^2 = 0^2 + 2(0.6)(5.85)$$

$$v = 2.65 \text{ m/s} \checkmark$$

Question 24:



$$T - 0.26g \sin \alpha = 0.26a$$

$$a = -10 \left(\frac{16}{65} \right)$$

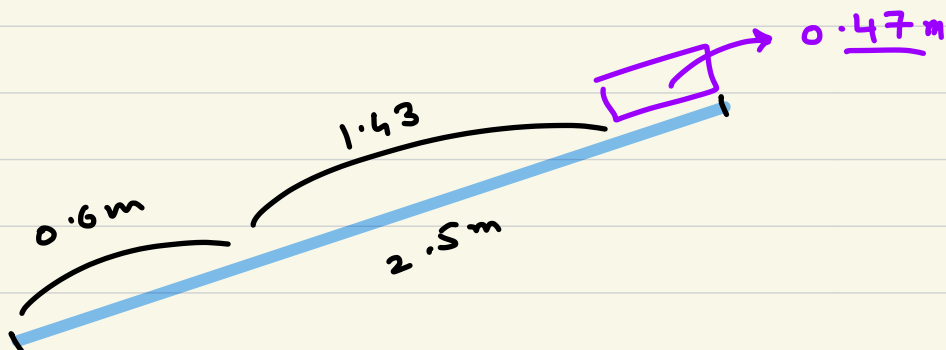
$$a = -2.46 \text{ m/s}^2$$

$$u = 2.65, \quad v = 0, \quad a = -2.46$$

$$0^2 = 2.65^2 + 2(-2.46)(s)$$

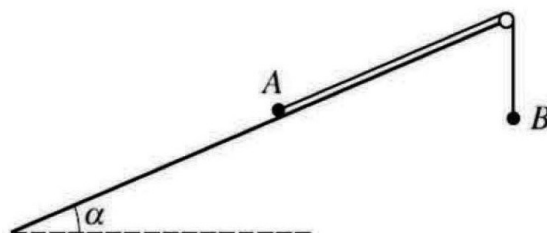
$$s = \cancel{1.19} \text{ m} \quad 1.43$$

$$0.60 + \cancel{1.19} = \cancel{1.79} \text{ m}$$



- 24 (i) $T - 0.26g(16 \div 65) = 0.26a$ or
 $0.52g - T = 0.52a$
For $\{0.52g - T = 0.52a$ or
 $T - 0.26g(16 \div 65) = 0.26a\}$
or $0.52g - 0.26g(16 \div 65) = (0.52 + 0.26)a$
Acceleration is 5.85 ms^{-2}
Tension is 2.16 N
- (ii) $[v^2 = 2 \times (76/13) \times 0.6]$
Speed is 2.65 ms^{-1}
 $0 = 91.2/13 - 2(160/65)s$
 $S = 57/40 \quad (= 1.425)$
 $[AP = 2.5 - 0.6 - 1.425]$
Distance AP is 0.475 m ✓

Question 23:



A light inextensible string has a particle A of mass 0.26 kg attached to one end and a particle B of mass 0.54 kg attached to the other end. The particle A is held at rest on a rough plane inclined at angle α to the horizontal, where $\sin \alpha = \frac{5}{13}$. The string is taut and parallel to a line of greatest slope of the plane. The string passes over a small smooth pulley at the top of the plane. Particle B hangs at rest vertically below the pulley (see diagram). The coefficient of friction between A and the plane is 0.2 . Particle A is released and the particles start to move.

- (i) Find the magnitude of the acceleration of the particles and the tension in the string. [6]

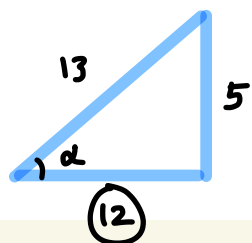
Particle A reaches the pulley 0.4 s after starting to move.

- (ii) Find the distance moved by each of the particles. [2]

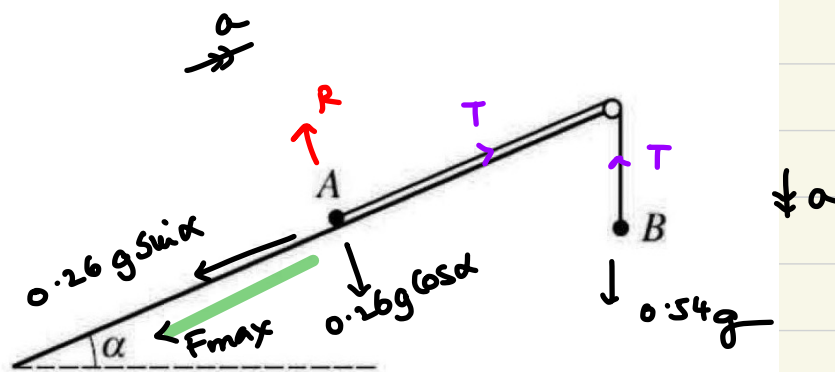
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Question 23:

$$\sin \alpha = \frac{5}{13}$$



$$\cos \alpha = \frac{12}{13}$$



$$\begin{aligned} R &= 0.26g \cos \alpha \\ &= 0.26(10)\left(\frac{12}{13}\right) \\ &= \end{aligned}$$

$$0.54g - T = 0.54a$$

$$T - 0.26g\left(\frac{5}{13}\right) - 0.48 = 0.26a$$

$$\begin{aligned} F_{\max} &= \mu R \\ &= 0.2(\quad) \\ &= 0.48 \text{ N.} \end{aligned}$$

$$a = 4.9 \text{ m/s}^2$$

$$u = 0, \quad a = 4.9, \quad t = 0.4 \text{ sec}$$

$$s = ut + \frac{1}{2}at^2$$

$$= \frac{1}{2}(4.9)(0.4)^2$$

$$= 0.392 \text{ m}$$

$$R = 2.6 \times (12 \div 13) (= 2.4)$$

$$[F = 0.2 \times 2.4]$$

$$[T - 2.6(5 \div 13) - F = 0.26a, 5.4 - T = 0.54a]$$

$$\text{For any two of } T - 1 - 0.48 = 0.26a, 5.4 - T = 0.54a \text{ or}$$

$$(5.4 - 1 - 0.48) = (0.54 + 0.26)a$$

$$\text{Acceleration is } 4.9 \text{ ms}^{-2}$$

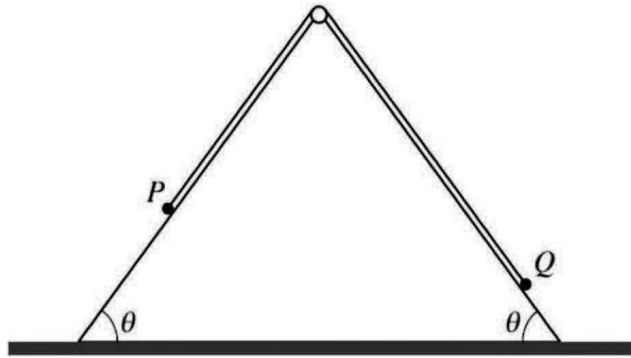
$$\text{Tension is } 2.75 \text{ N (2.754 exact)}$$

$$[s = \frac{1}{2} 4.9 \times 0.4^2]$$

$$\text{Distance is } 0.392 \text{ m}$$



Question 18:



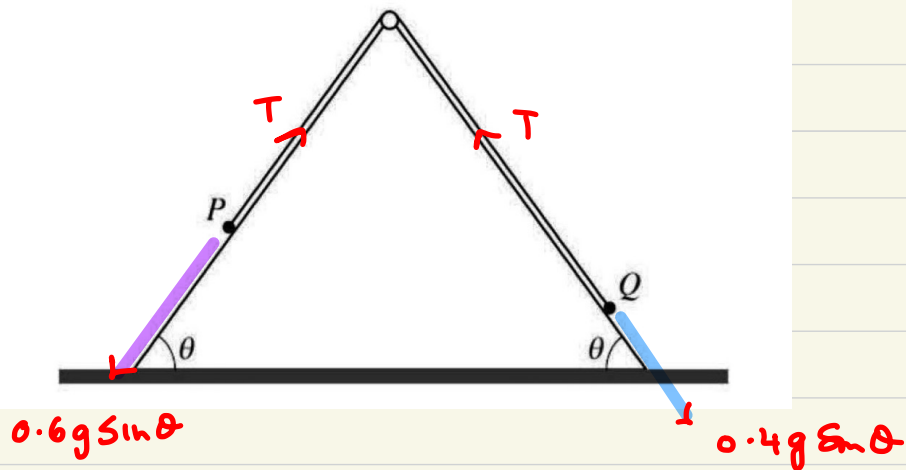
Particles P and Q , of masses 0.6 kg and 0.4 kg respectively, are attached to the ends of a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a vertical cross-section of a triangular prism. The base of the prism is fixed on horizontal ground and each of the sloping sides is smooth. Each sloping side makes an angle θ with the ground, where $\sin \theta = 0.8$. Initially the particles are held at rest on the sloping sides, with the string taut (see diagram). The particles are released and move along lines of greatest slope.

- (i) Find the tension in the string and the acceleration of the particles while both are moving. [5]

The speed of P when it reaches the ground is 2 ms^{-1} . On reaching the ground P comes to rest and remains at rest. Q continues to move up the slope but does not reach the pulley.

- (ii) Find the time taken from the instant that the particles are released until Q reaches its greatest height above the ground. [4]

Question 18:



$$0.6g \sin \theta - T = 0.6a$$

$$\checkmark \quad \textcircled{T} - 0.4g \sin \theta = 0.4a$$

$$0.2(10)(0.8) = 1.0a$$

$$\therefore a = 1.6 \text{ m/s}^2$$

$$T =$$

$$u = 0, \quad a = 1.6$$

$$v = 2 \text{ m/s}$$

Time

dist.

$$v = u + at$$

$$2 = 0 + 1.6(t) \quad \therefore t = 1.25 \text{ sec (system)}$$

$$\cancel{T} - \boxed{0.4}g \sin \theta = \boxed{0.4}a$$

$$-(10)(0.8) = a \quad \therefore a = -8 \text{ m/s}^2$$

$$u = +2, \quad a = -8, \quad v = 0 \Rightarrow v = u + at$$

$$0 = 2 + (-8)t$$

$$t = 0.25 \text{ sec}$$

$$\text{Total Time } 1.25 + 0.25 = 1.50 \text{ sec}$$

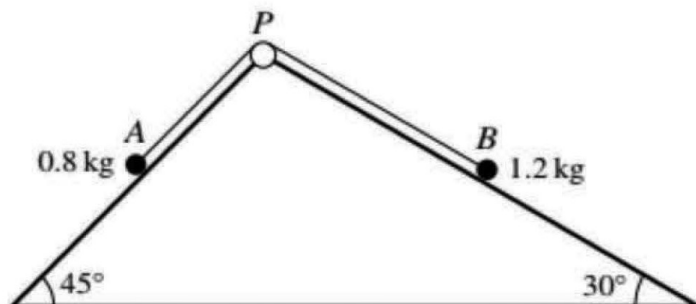
free

- 18 (i) $0.6g \times 0.8 - T = 0.6a$ and $T - 0.4g \times 0.8 = 0.4a$
or $(0.6 - 0.4)g \times 0.8 = (0.6 + 0.4)a$
Tension is 3.84 N or acceleration is 1.6ms^{-2}
Acceleration is 1.6ms^{-2} or tension is 3.84 N
- (ii) $2 = 1.6t_1$ $(t_1 = 1.25)$
 $0 = 2 - 0.8gt_2$ $(t_2 = 0.25)$
Time taken in 1.5 s



Lec '05

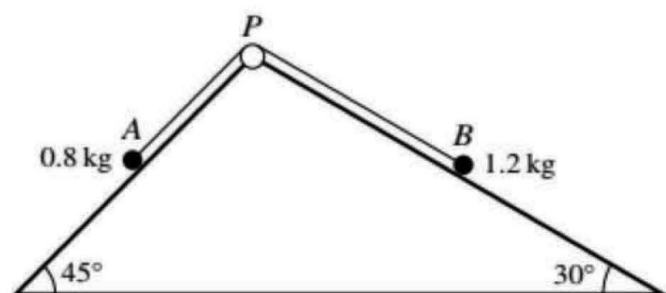
Question 52:



The diagram shows a triangular block with sloping faces inclined to the horizontal at 45° and 30° . Particle A of mass 0.8 kg lies on the face inclined at 45° and particle B of mass 1.2 kg lies on the face inclined at 30° . The particles are connected by a light inextensible string which passes over a small smooth pulley P fixed at the top of the faces. The parts AP and BP of the string are parallel to lines of greatest slope of the respective faces. The particles are released from rest with both parts of the string taut. In the subsequent motion neither particle reaches the pulley and neither particle reaches the bottom of a face.

- (i) Given that both faces are smooth, find the speed of A after each particle has travelled a distance of 0.4 m . [6]
- (ii) It is given instead that both faces are rough. The coefficient of friction between each particle and a face of the block is μ . Find the value of μ for which the system is in limiting equilibrium. [6]

Question 52:



A52.

7(i)

$$A \quad T - 0.8g \sin 45 = 0.8a$$

$$B \quad 1.2g \sin 30 - T = 1.2a$$

$$\text{System} \quad 1.2g \sin 30 - 0.8g \sin 45 = 2a$$

$$a = 0.171$$

$$v^2 = 2 \times a \times 0.4$$

$$v = 0.370 \text{ so speed of } A \text{ is } 0.370 \text{ ms}^{-1}$$

Alternative scheme for Question 7(i)

$$\text{KE gain} = \frac{1}{2} \times 0.8 \times v^2 + \frac{1}{2} \times 1.2 \times v^2$$

PE loss =

$$1.2g \times 0.4 \sin 30 - 0.8g \times 0.4 \sin 45$$

$$\frac{1}{2} \times 0.8 \times v^2 + \frac{1}{2} \times 1.2 \times v^2 =$$

$$1.2g \times 0.4 \sin 30 - 0.8g \times 0.4 \sin 45$$

$$v = 0.370 \text{ so speed of } A \text{ is } 0.370 \text{ ms}^{-1}$$

7(ii)

$$R_A = 0.8g \cos 45 = 4\sqrt{2}$$

$$R_B = 1.2g \cos 30 = 6\sqrt{3}$$

$$F_A = 4\sqrt{2} \mu \text{ and } F_B = 6\sqrt{3} \mu$$

$$A \quad 0.8g \sin 45 + F_A = T$$

$$B \quad 1.2g \sin 30 - F_B = T$$

or system equation:

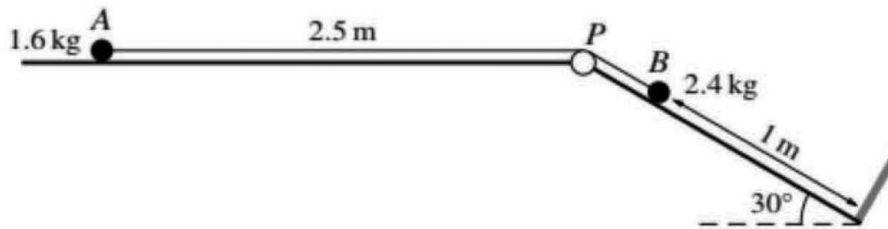
$$12 \sin 30 - 8 \sin 45 = F_A + F_B$$

Correct equation(s)

$$\mu = \frac{(6 - 4\sqrt{2})}{(6\sqrt{3} + 4\sqrt{2})}$$

$$= 0.0214$$

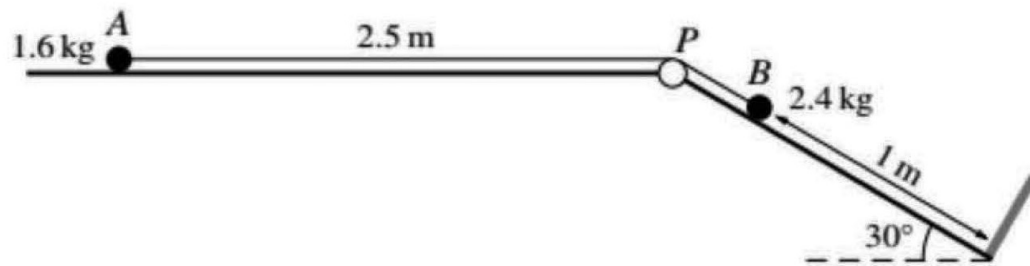
Question 53:



As shown in the diagram, a particle A of mass 1.6 kg lies on a horizontal plane and a particle B of mass 2.4 kg lies on a plane inclined at an angle of 30° to the horizontal. The particles are connected by a light inextensible string which passes over a small smooth pulley P fixed at the top of the inclined plane. The distance AP is 2.5 m and the distance of B from the bottom of the inclined plane is 1 m . There is a barrier at the bottom of the inclined plane preventing any further motion of B . The part BP of the string is parallel to a line of greatest slope of the inclined plane. The particles are released from rest with both parts of the string taut.

- (i) Given that both planes are smooth, find the acceleration of A and the tension in the string. [5]
- (ii) It is given instead that the horizontal plane is rough and that the coefficient of friction between A and the horizontal plane is 0.2 . The inclined plane is smooth. Find the total distance travelled by A . [9]

Question 53:



A53.

7(i)

$$[T = 1.6a, 2.4g \sin 30 - T = 2.4a]$$
$$\text{System is } 2.4g \sin 30 = 4a$$

$$a = 3$$

$$T = 4.8$$

7(ii)

Friction force on A is
 $F = 0.2 \times 1.6g [= 3.2]$

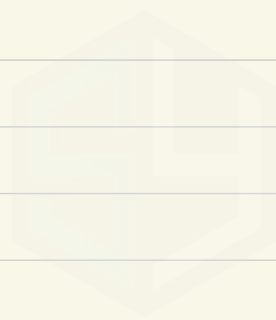
$$T - F = 1.6a$$

$$2.4g \sin 30 - T = 2.4a$$

$$\text{System is } 2.4g \sin 30 - F = 4a$$

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TOPIC 03: NEWTONS LAWS OF MOTION (B) Others



Shahab Yakoob

Question 2

S_1 and S_2 are light inextensible strings, and A and B are particles each of mass 0.2 kg . Particle A is suspended from a fixed point O by the string S_1 , and particle B is suspended from A by the string S_2 . The particles hang in equilibrium as shown in the diagram.

- (i) Find the tensions in S_1 and S_2 . [3]

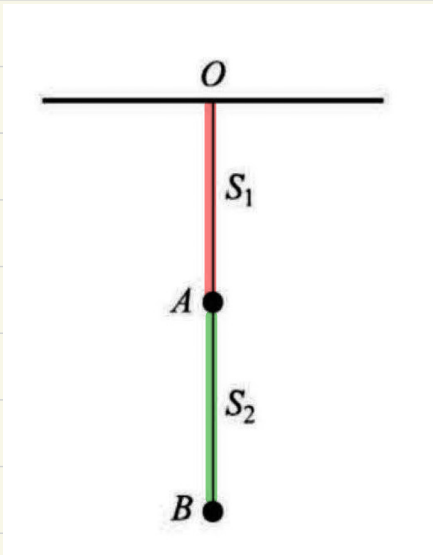
The string S_1 is cut and the particles fall. The air resistance acting on A is 0.4 N and the air resistance acting on B is 0.2 N .

- (ii) Find the acceleration of the particles and the tension in S_2 . [5]

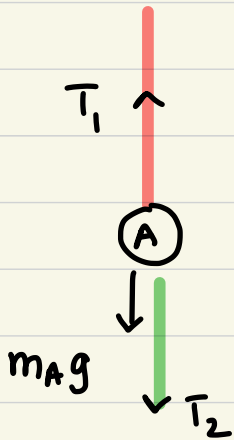
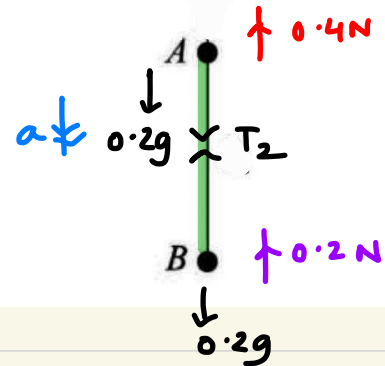
June 2003 Paper 4 Q5

i, EQUILIBRIUM

$$m_A = m_B = 0.2 \text{ kg}$$



UNIQUE CASE $\rightarrow$ Connected Particles



$$A : T_2 + 0.2g - 0.4 = 0.2a$$

$$B : 0.2g - T_2 - 0.2 = 0.2a$$

$$\text{system} \quad 2 + 2 - 0.6 = 0.4a$$

$$3.4 = 0.4a$$

$$\therefore a = 8.5 \text{ m/s}^2$$

$$T_1 = 0.2g + T_2$$

$$= 2 + 2$$

$$T_1 = 4 \text{ N}$$

$$T_2 = 0.2(10)$$

$$= 2 \text{ N}$$

$$T_2 + 2 - 0.4 = 0.2(8.5)$$

$$T_2 =$$

- 2 (i) For resolving forces on any two of A , or B , or A and B combined ($T = W_A + T_2, T_2 = W_B, T_1 = W_A + W_B$)

Tension in S_1 is 4 N or Tension in S_2 is 2 N
Accept $0.4g$ or 3.92 (from 9.8 or 9.81) for T_1
Tension in S_2 is 2 N or Tension in S_1 is 4 N
Accept $0.2g$ or 1.96 (from 9.8 or 9.81) for T_2

SR (for candidates who omit g) (Max 1 out of 3)
 $T_1 = 0.4$ and $T_2 = 0.2$ B1

- (ii) For applying Newton's second law to A , or to B , or to A and B combined

For any one of the equations $T + 2 - 0.4 = 0.2a$,
 $2 - T - 0.2 = 0.2a, 4 - 0.4 - 0.2 = 0.4a$

For a second of the above equations

For solving the simultaneous equations for a and T

Acceleration is 8.5 ms^{-2} , tension is 0.1 N
Accept 8.3 from 9.8 or 8.31 from 9.81

SR (for candidates who obtain only the 'combined' equation) (Max 3 out of 5)

For applying Newton's second law to A and B combined

For $4 - 0.4 - 0.2 = 0.4a$
Acceleration is 8.5 ms^{-2}

Question 3

A small block of mass 0.15 kg moves on a horizontal surface. The coefficient of friction between the block and the surface is 0.025 .

(i) Find the frictional force acting on the block.

(ii) Show that the deceleration of the block is 0.25 m s^{-2} .

$$0 - 0.0375 = 0.15a \quad \therefore a = -0.25 \text{ m/s}^2 = 0.0375 \text{ N}$$

The block is struck from a point A on the surface and, 4 s later, it hits a boundary board at a point B. The initial speed of the block is 5.5 m s^{-1} .

(iii) Find the distance AB.

The block rebounds from the board with a speed of 3.5 m s^{-1} and moves along the line BA. Find

(iv) the speed with which the block passes through A,

(v) the total distance moved by the block, from the instant when it was struck at A until the instant when it comes to rest.

June 2003 Paper 4 Q6

A

B

$$\begin{aligned} \text{(ii)} \quad u &= 5.5 \text{ m/s} \\ a &= -0.25 \text{ m/s}^2 \\ t &= 4 \text{ sec} \end{aligned}$$

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ &= 5.5(4) + \frac{1}{2}(-0.25)(4)^2 \end{aligned}$$

$$s = 20 \text{ m}$$

A ← B

$$\begin{aligned} \text{(iv)} \quad u &= 3.5 \text{ m/s} \\ a &= -0.25 \text{ m/s}^2 \end{aligned}$$

$$s = 20$$

$$v^2 = 3.5^2 + 2(-0.25)(20)$$

$$v = 1.5 \text{ m/s}$$

ALT

$$\begin{aligned} u &= 1.5 \\ a &= -0.25 \\ v &= 0 \end{aligned}$$

$$\begin{aligned} v^2 &= u^2 + 2as \\ s &= 4.5 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad u &= 3.5, \quad v = 0 \\ a &= -0.25 \\ s &= ? \end{aligned}$$

$$\begin{aligned} 0^2 &= 3.5^2 + 2(-0.25)s \\ s &= 24.5 \text{ m} \end{aligned}$$

$$\text{Total Dist} = (20 + 24.5) \text{ m}$$

- 3 (i) For using $F = \mu R$ and $R = mg$ ($F = 0.025 \times 0.15 \times 10$)

Frictional force is 0.0375 N or 3/80 N
Accept 0.0368 from 9.8 or 9.81

- (ii) For using $F = ma$ ($-0.0375 = 0.15a$) or $d = \mu g$

Deceleration is 0.25 ms^{-2} (or $a = -0.25$) A.G.

- (iii) For using $s = ut + \frac{1}{2}at^2$ ($s = 5.5 \times 4 + \frac{1}{2}(-0.25)16$)

Distance AB is 20m

- (iv) For using $v^2 = u^2 + 2as$ ($v^2 = 3.5^2 - 2 \times 0.25 \times 20$)

Speed is 1.5 ms^{-1} (ft $\sqrt{(24.5 - (iii)) / 2}$)

- (v) Return dist. = $\frac{3.5^2}{2 \times 0.25}$ or distance beyond A = $\frac{(iv)^2}{2 \times 0.25}$

Total distance is 44.5 m
(ft $24.5 + (iii)$ or $2((iv)^2 + (iii))$)

Question 6

A particle of mass m kg moves up a line of greatest slope of a rough plane inclined at 21° to the horizontal. The frictional and normal components of the contact force on the particle have magnitudes F N and R N respectively. The particle passes through the point P with speed 10 m s^{-1} , and 2 s later it reaches its highest point on the plane.

(i) Show that $R = 9.336m$ and $F = 1.416m$, each correct to 4 significant figures. [5]

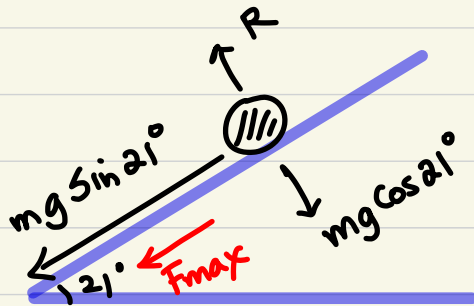
(ii) Find the coefficient of friction between the particle and the plane. [1]

After the particle reaches its highest point it starts to move down the plane.

(iii) Find the speed with which the particle returns to P . [5]

Nov 2006 Paper 4 Q7

(i),



$$\left. \begin{array}{l} u = 10 \\ t = 2 \\ v = 0 \end{array} \right\} \begin{array}{l} v = u + at \\ 0 = 10 + (a)(2) \\ \therefore a = -5 \text{ m/s}^2 \end{array}$$

$$\begin{aligned} \checkmark \quad R &= mg \cos 21^\circ \\ R &= m(9.336) \end{aligned}$$

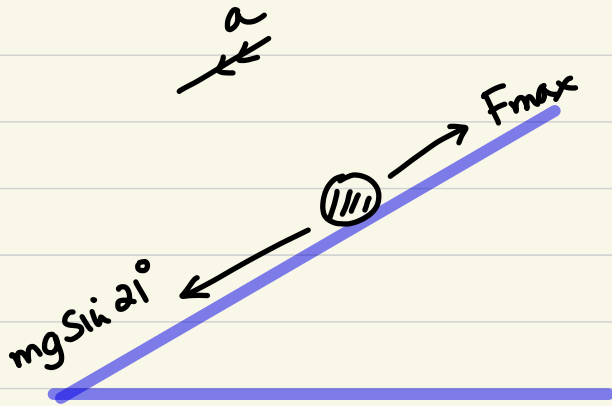
$$\begin{aligned} \checkmark \quad 0 &= m(10)(0.358) - F_{\max} \\ &= m(-5) \\ -3.584m - F_{\max} &= -5m \end{aligned}$$

$$5m - 3.584m = F_{\max}$$

(ii), $F_{\max} = \mu R$

$$F_{\max} = 1.416m$$

$$\mu = \frac{F_{\max}}{R} = \frac{1.416}{9.336} \text{ or } 0.152$$



$$R = 9.336 \text{ m}$$

$$F_{\max} = 1.416 \text{ m}$$

$$m(3.584) - 1.416 \text{ m} = m a$$

$$a = 2.167 \cdot \text{m/s}^2$$

$$u = 10$$

$$t = 2$$

$$v = 0$$

$$a = -5$$

$$v^2 = u^2 + 2as$$

$$0^2 = 10^2 + 2(-5)s$$

$$s = 10 \text{ m}$$

$$u = 0$$

$$a = 2.167$$

$$s = 10$$

$$v^2 = 0^2 + 2(2.167)(10)$$

$$v =$$



$$6 \text{ (i) } R = mg \cos 21^\circ = 9.336m$$

$$a = -5$$

$$[-mg \sin 21^\circ - F = -5m]$$

$$F = 1.416m$$

$$\text{(ii) Coefficient is } 0.152$$

$$\text{(iii) } s = 10$$

$$mg \sin 21^\circ - 1.416m = ma \quad [a = 2.167 \text{ m/s}^2]$$

$$[v^2 = 2(g \sin 21^\circ - 1.416)10]$$

Question 12

A particle P of mass 0.2 kg is released from rest at a point 7.2 m above the surface of the liquid in a container. P falls through the air and into the liquid. There is no air resistance and there is no instantaneous change of speed as P enters the liquid. When P is at a distance of 0.8 m below the surface of the liquid, P 's speed is 6 m s^{-1} . The only force on P due to the liquid is a constant resistance to motion of magnitude $R \text{ N}$.

(i) Find the deceleration of P while it is falling through the liquid, and hence find the value of R .

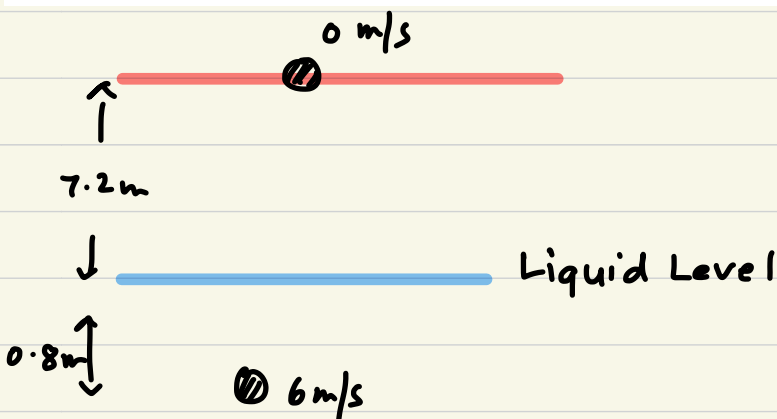
[5]

The depth of the liquid in the container is 3.6 m . P is taken from the container and attached to one end of a light inextensible string. P is placed at the bottom of the container and then pulled vertically upwards with constant acceleration. The resistance to motion of $R \text{ N}$ continues to act. The particle reaches the surface 4 s after leaving the bottom of the container.

(ii) Find the tension in the string.

[4]

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$$u = 0, a = 10 \text{ m/s}^2, s = 7.2 \text{ m}$$

$$v^2 = 0^2 + 2(10)(7.2)$$

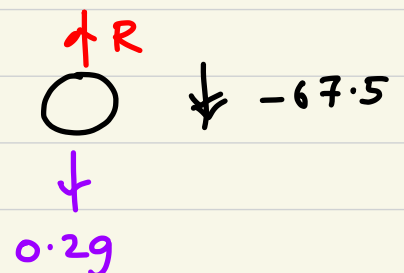
$$v = 12 \text{ m/s} \checkmark$$

$$u = 12, v = 6, s = 0.8$$

$$6^2 = 12^2 + 2(0.8)(a)$$

$$\frac{36 - 144}{1.6} = a$$

$$\therefore a = -67.5 \text{ m/s}^2$$



$$0.2(10) - R = 0.2(-67.5)$$

$$2 - R = -13.5$$

$$\therefore R = 15.5 \text{ N}$$

Question 12

A particle P of mass 0.2 kg is released from rest at a point 7.2 m above the surface of the liquid in a container. P falls through the air and into the liquid. There is no air resistance and there is no instantaneous change of speed as P enters the liquid. When P is at a distance of 0.8 m below the surface of the liquid, P 's speed is 6 m s^{-1} . The only force on P due to the liquid is a constant resistance to motion of magnitude $R \text{ N}$.

(i) Find the deceleration of P while it is falling through the liquid, and hence find the value of R .

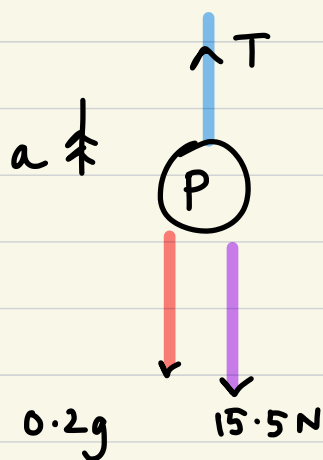
[5]

The depth of the liquid in the container is 3.6 m . P is taken from the container and attached to one end of a light inextensible string. P is placed at the bottom of the container and then pulled vertically upwards with constant acceleration. The resistance to motion of $R \text{ N}$ continues to act. The particle reaches the surface 4 s after leaving the bottom of the container.

(ii) Find the tension in the string.

[4]

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$$u = 0$$

$$t = 4 \text{ s}$$

$$s = 3.6 \text{ m}$$

$$s = ut + \frac{1}{2}at^2$$

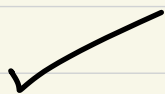
$$3.6 = \frac{1}{2}(a)(4)^2$$

$$a = \frac{7.2}{16} \text{ or } 0.45 \text{ m/s}^2$$

$$T - 0.2g - 15.5 = 0.2(0.45)$$

$$T - 2 - 15.5 = 0.2(0.45)$$

$$T = 17.6 \text{ N}$$



Lec # 06

"Laws of Motion"

The End!

12

- (i) $v^2 = 2 \times g \times 7.2$
 $\rightarrow$ speed at surface is 12 ms^{-1}

$$[6^2 = 12^2 + 2a \times 0.8]$$

Deceleration is 67.5 ms^{-2}

$$[0.2g - R = -0.2 \times 67.5]$$

$$R = 15.5$$

$$[3.6 = \frac{1}{2} a \times 4^2]$$

$$a = 0.45 \text{ ms}^{-2}$$

$$[T - R - 0.2g = 0.2 \times 0.45]$$

Tension is 17.6 N (17.59 exact)

Alternative Energy Method

(i) $0.2g \times 8 = R(0.8) + \frac{1}{2} (0.2) 6^2$

$$R = 15.5$$

$$0.2g - 15.5 = 0.2a$$

$$a = -67.5$$

- (ii)

$$3.6 = v/2 \times 4 \quad v = 1.8$$

$$T(3.6) = R(3.6) + 0.2g(3.6) + \frac{1}{2}(0.2)1.8^2$$

$$T = 17.6 \text{ N}$$